



Growing Patterns

Quick facilitation notes & answer key

Goal

Children read a three-part (ABC) repeating pattern, extend it by one and by two shapes, and create their own. Extending a pattern by more than one step builds prediction — a computational-thinking skill behind repeating a process.

Kindergarten frame (Ontario)

Problem Solving and Innovating — children read, extend, and create repeating patterns.

No reading required of the child: read each prompt aloud; the child draws the shapes.

Materials

The printed worksheet + a pencil/crayon. Optional: pattern blocks or coloured beads to build ABC patterns by hand.

Run it (about 15 min)

1. MODEL: say the three-part pattern aloud — 'triangle, square, circle...'. The next is a triangle.
2. GUIDED: do Exercise 1 together — ● ■ ★ repeating, the next is ●.
3. YOU TRY: children predict the next TWO in Exercise 2 (● then ▲), then create their own in Exercise 3.
4. Connect: 'Computers use patterns to repeat steps again and again — just like your pattern keeps going.'

Answer key (modelled by the run-gate)

Model — ▲ ■ ● ▲ ■ ● → next is ▲.

Ex 1 — ● ■ ★ ● ■ ★ → next is ●.

Ex 2 — ▲ ● ▲ ● ▲ → next two are ● then ▲.

Ex 3 — any pattern that truly repeats is correct (e.g. ★ ● ★ ● ..., or ● ▲ ■ ● ▲ ■).

Watch for & differentiate

Common slip: in Ex 2, drawing only one shape — point to both dashed boxes.

Simplify: do two-part patterns first, then add the third shape.

Extend: ask the child to read their own pattern aloud and continue it for two more shapes.

Movement option: stomp-clap-jump, stomp-clap-jump — what three actions come next?

Success looks like

The child extends a three-part pattern correctly, draws two correct shapes in Exercise 2, and creates a pattern in Exercise 3 that genuinely repeats.